

University College London
Department of Economics
Main Summer Assessment Period 2025
ECON0039
Advanced Macroeconomics

Student Name:

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Student Number:

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Time allowance: You have **TWO (2) hours** to complete this examination.

If you have been granted SoRA extra time and/or rest breaks, your individual examination duration will be extended pro-rata.

Calculators (UCL approved models only) **are** permitted in this examination.

Number of Questions Answered Policy: In cases where a student answers more questions than requested by the examination rubric, the policy of the Department of Economics is that the student's first set of answers up to the required number will be the ones that count (not the best answers). All remaining answers will be ignored.

Recording of answers:

- Use the separate paper provided to do all the necessary rough work, calculations or drafting of answers. If you need more paper for rough work, alert the invigilators by raising your hand.
- Enter your final answers, answer steps, and explanation/interpretation (as required) in the answer-boxes provided within this question-and-answer booklet.
- **Only results, workings, diagrams, and writing provided within the appropriate boxed answer-spaces will be marked.** Comments, notes or anything else outside of the provided boxed answer-spaces will be ignored.
- **Do not** remove any pages from this answer booklet!
- If you inadvertently damage your answer booklet during the exam, alert the invigilators and request a new answer booklet. You will **not** get additional time!

Submission of answers:

- Ensure that you have provided in the spaces above your **correct student number** (see your student card) and your **correct name**, in block capitals. Failure to provide this information may mean that any marks awarded cannot be allocated to you correctly.
- Only submit your answer booklet! Submitting a damaged or incomplete answer booklet may result in a mark of 0.

The remainder of this page has been left blank intentionally. Do not use the space on this page to record your answers to questions in this exam.

Exam Instructions and Questions

You must answer ALL questions from those presented in the remainder of this booklet. Specifically, you must answer TWO questions from Part A, FOUR questions from Part B and THREE questions from PART C.

Marks for each question are specified in the instructions for that part.

The remainder of this page has been left blank intentionally. Do not use the space on this page to record your answers to questions in this exam.

Part A: True/False/Uncertain Questions (30 points)

Answer ALL questions from Part A by specifying whether the statement is TRUE/FALSE/-UNCERTAIN. For each question, write a concise explanation and include equations and graph(s) if needed (you can use formulas and diagrams, but you MUST write your argument in words too). Make sure to clearly mention on which theory/model your argument is based: maximum marks will be given for a correct explanation that explicitly reference the course material. Each question is worth 15 points. Record your answers in the space below the questions. If you need more space, use the ADDITIONAL SPACE at the end of the booklet, indicating clearly to which question(s) your answer pertains.

A1. There was no growth of total income during the Malthusian regime.

Record your answer to Question A1 in the space below. Do NOT use this space to record answers to any questions other than A1.

–Solution– FALSE

1. The Malthusian regime is defined as the economic phase that followed the hunter-gatherers societies and the invention of agriculture and preceded the industrial revolution.
2. Malthus was Professor of Economy. He wrote in 1798 *An Essay on the Principle of Population*. He argued that population growth generally expanded in times and in regions of plenty until the size of the population relative to the primary resources caused distress. Two types of checks hold population within resource limits: positive checks, which raise the death rate (*hunger, disease and war*) and preventive checks, which lower the birth rate (*abortion, birth control, prostitution, postponement of marriage and celibacy*).
3. The Malthusian regime is characterized by growth in total income and population, but stagnation of income per capita. Any changes in the availability of resources (for example some technical progress) translates in an increase in population and total income, but no change in income per capita, that stays at the subsistence level.
4. In the Malthusian regime, comparative statics is as follows: technical progress increases population, not income per capita; death rate increases income per capita (example of the Black Death in England, 1348), fertility rate decreases income per capita.
5. A model that can represent that regime is given by the three equations: (1) the birth rate b is high when income per capita y is high: $b = b(y)$, $b' > 0$; (2) the mortality rate m is low when income per capita y is high: $m = m(y)$, $m' < 0$; (3) population P is low when income per capita y is high: $P = P(y)$, $P' < 0$. (you can plot these three equations and show what happens when there is a increase in productivity)

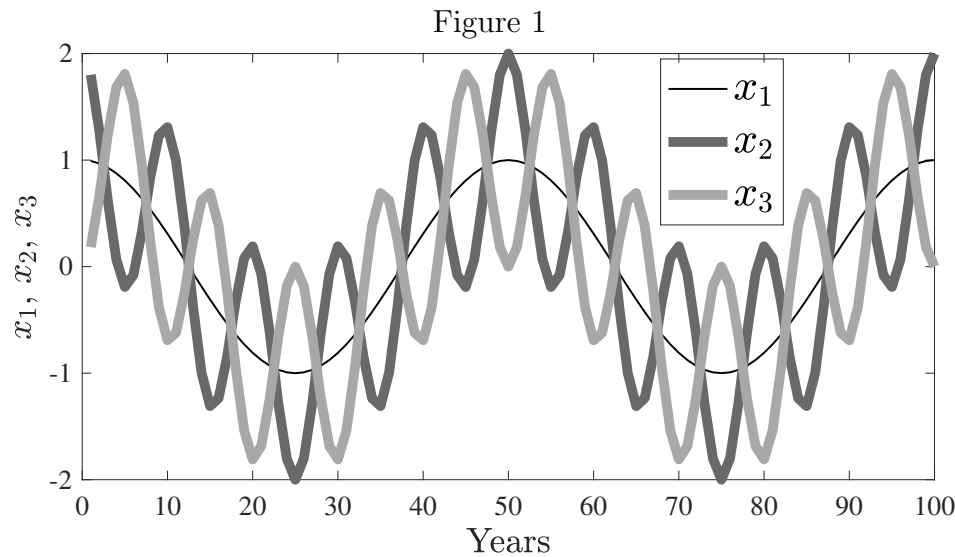
PART A cont'd: Answer ALL questions from Part A

Additional answer-space for Question A1.

–Solution– See previous page.

PART A cont'd: Answer ALL questions from Part A

A2. Consider Figure 1 below. Are the series x_2 and x_3 negatively correlated? Relate your answer to what you know about time series filtering and analysis in the frequency domain.



Record your answer to Question A2 in the space below. Do NOT use this space to record answers to any questions other than A2.

–Solution– UNCERTAIN

- We see that x_2 and x_3 feature two types of oscillations.
- At low frequency (periodicity of 50 years), they move with x_1 : x_2 and x_3 are decreasing from year 1 to year 25, then increasing from year 25 to year 50, etc... At that frequency, there are positively correlated.
- At high frequency (periodicity of 10 years), x_2 and x_3 move in opposite direction: one is high when the other is low. At that frequency, there are negatively correlated.
- Depending on which frequency we are focusing on (in other words depending how we filter the two series, we would find different correlations.
- If we compute the correlation between x_2 and x_3 , we find 0, but it is the consequence of two different phenomena: very positive correlation at low frequencies and very negative one at high frequencies.
- See next page.

Additional answer-space for Question A2.

–Solution–

- One way to filter x_2 and x_3 could be:

- Compute the high frequency versions of x_2 and x_3 as

$$x_2^{HF} = x_2 - x_1 \quad \text{and} \quad x_3^{HF} = x_3 - x_1.$$

Correlation between x_2^{HF} and x_3^{HF} is highly negative (it is actually -1).

- Compute the low frequency versions of x_2 and x_3 as

$$x_2^{LF} = x_2 - x_2^{HF} \quad \text{and} \quad x_3^{LF} = x_3 - x_3^{HF}.$$

Obviously, $x_2^{LF} = x_1$ and $x_3^{LF} = x_1$, so that the correlation between x_2^{LF} and x_3^{LF} is highly positive (it is actually 1).

PART B: Problem (40 points)

Answer ALL questions of the following Part B. Question B1 is worth 4 points and the other questions are worth 12 points. Write your answers in words and include equations if needed or specified by the question. Record your answer to each question in the space below the question. If you need more space, use the **ADDITIONAL SPACE** at the end of the booklet, indicating clearly to which question(s) your answer pertains.

This problem considers a monetary model in the UV setting of Lecture 2 on Business Cycles. The economy features a representative firm, a representative household and a Central Bank. The representative household owns the firm and receives its profits π . There is a unique produced good that can be used for consumption c and investment x . Its price is p . Money is the numéraire good. It is supplied in quantity $\bar{m}$ by the Central Bank and is distributed to the households.

The representative household utility is of the UV form, and is given by

$$U\left(c, \ell, \frac{m}{p}\right) + V(x, \theta) = \ln c - B\ell + \nu \ln\left(\frac{m}{p}\right) + \theta \ln x,$$

where m is the money demand of the household, ℓ is labour. Agents demand money because $\frac{m}{p}$ brings some utility.

The household budget constraint is

$$pc + px + m \leq w\ell + \bar{m} + \pi$$

where w is the nominal wage.

The representative firm produces a single good according to the technology $y = A\ell^\alpha$ with $0 < \alpha < 1$. y is split into consumption c and investment x .

The Central Bank follows the money supply rule:

$$\bar{m} = \mu - \varphi w,$$

where parameters μ and φ are such that $\bar{m}$ is strictly positive in equilibrium. Finally, all markets are competitive, prices are flexible and all parameters (A, B, ν, θ, μ and φ) are strictly positive.

PART B cont'd: Answer ALL questions from PART B

B1. Write the profit maximization problem of the firm and the first order conditions to this maximization.

Record your answer to Question B1 in the space below. Do NOT use this space to record answers to any questions other than B1.

–**Solution**– The representative firm solves the problem:

$$\max_{\ell} \pi = py - w\ell = pA\ell^{\alpha} - w\ell,$$

and first order condition is

$$\alpha pA\ell^{\alpha-1} = w.$$

PART B cont'd: Answer ALL questions from PART B

B2. Write the utility maximization problem of the household and the four first order conditions to this maximization, with respect to c , x , ℓ and m . Derive consumption demand c , investment demand x , money demand m^d and labour supply ℓ^s as a function of w , p , π , $\bar{m}$ and parameters.

Record your answer to Question B2 in the space below. Do NOT use this space to record answers to any questions other than B2.

–**Solution**– The representative household solves the problem:

$$\max_{c, \ell, m, x} \ln c - B\ell + \nu \ln \left(\frac{m}{p} \right) + \theta \ln x,$$

subject to the budget constraint

$$pc + px + m \leq w\ell + \bar{m} + \pi \quad (\lambda).$$

First order conditions are

$$\frac{1}{c} = \lambda p \quad (1)$$

$$B = \lambda w \quad (2)$$

$$\frac{\nu}{m} = \lambda \quad (3)$$

$$\frac{\theta}{x} = \lambda p \quad (4)$$

(1) and (3) give $pc = \frac{m}{\nu}$. (1) and (4) imply $x = \theta c$. Putting all together, we obtain $c = \frac{1}{B} \frac{w}{p}$, $m^d = \nu \frac{w}{B}$ and $x = \frac{\theta}{B} \frac{w}{p}$. Total good demand is then $c + x = \frac{1+\theta}{B} \frac{w}{p}$. Plugging those three expressions in the budget constraint, we obtain $\ell^s = \frac{(1+\theta+\nu)}{B} - \frac{(\bar{m}+\pi)}{w}$.

PART B cont'd: Answer ALL questions from PART B

Additional answer-space for Question B2.

–Solution– See previous page.

PART B cont'd: Answer ALL questions from PART B

B3. Define the general equilibrium of this economy. From money market equilibrium condition, find equilibrium nominal wage w^E . Find the equilibrium levels of quantities c^E , ℓ^E , x^E and y^E and price p^E .

Record your answer to Question B3 in the space below. Do NOT use this space to record answers to any questions other than B3.

–Solution–

A general equilibrium is a collection of prices and quantities such that, (i) given wage w^E and price p^E , quantities c^E , ℓ^E , x^E , y^E and m^E maximise utility and profits, (ii) the Central Bank follows the money policy rule, (iii) good, labour and money markets clear.

The table below gives the expressions for demand and supply on the three markets, as obtained above.

	Supply	Demand
Good market:	$y = A^{\frac{1}{1-\alpha}} \left(\frac{1}{\alpha} \frac{w}{p} \right)^{\frac{-\alpha}{1-\alpha}}$	$c + x = \frac{(1+\theta)}{B} \frac{w}{p}$
Labour market:	$\ell^s = \frac{(1+\theta+\nu)}{B} - \frac{(\bar{m}+\pi)}{w}$	$\ell^d = \left(\frac{1}{\alpha A} \frac{w}{p} \right)^{\frac{-1}{1-\alpha}}$
Money market:	$\bar{m} = \mu - \varphi w$	$m^d = \nu \frac{w}{B}$

where $\pi = (1 - \alpha) \alpha^{\frac{\alpha}{1-\alpha}} A^{\frac{1}{1-\alpha}} p^{\frac{1}{1-\alpha}} w^{\frac{-\alpha}{1-\alpha}}$.

Money market equilibrium implies $w^E = \left(\frac{\nu}{B} + \varphi \right)^{-1} \mu$

To solve for the equilibrium, $y = c + x$ implies $\frac{w^E}{p^E} = \alpha^\alpha A \left(\frac{B}{1+\theta} \right)^{1-\alpha}$.

Then labour demand gives $\ell^E = \alpha \frac{1+\theta}{B}$. Using the demands c and x , one gets $c^E = \alpha^\alpha A \frac{B^{-\alpha}}{(1+\theta)^{1-\alpha}}$ and $x^E = \theta \alpha^\alpha A \frac{B^{-\alpha}}{(1+\theta)^{1-\alpha}}$. From the production function, the expression for ℓ^E implies $y^E = \alpha^\alpha A \left(\frac{1+\theta}{B} \right)^\alpha$.

Finally, $p^E = \left(\frac{\nu}{B} + \varphi \right)^{-1} \frac{\mu}{\alpha^\alpha A} \left(\frac{(1+\theta)}{B} \right)^{(1-\alpha)}$.

PART B cont'd: Answer ALL questions from PART B

Additional answer-space for Question B3.

–Solution– See previous page.

PART B cont'd: Answer ALL questions from PART B

B4. What are the signs of the derivatives $M_{yA} = \frac{\partial y^E}{\partial A}$, $M_{pA} = \frac{\partial p^E}{\partial A}$, $M_{y\theta} = \frac{\partial y^E}{\partial \theta}$ and $M_{p\theta} = \frac{\partial p^E}{\partial \theta}$? Why does it make sense to label ∂A as a supply shock and $\partial \theta$ as a demand shock? What is the effect of the monetary policy parameter φ on the derivatives absolute values $|M|$? Discuss.

Record your answer to Question B4 in the space below. Do NOT use this space to record answers to any questions other than B4.

–Solution–

- It is easy to check that shocks have the following effect:

	∂y^E	∂p^E
$\partial A > 0$	+	-
$\partial \theta > 0$	+	+

- ∂A can be coined “supply shock” because it moves prices and quantities in opposite direction, while $\partial \theta$ can be coined “demand shock” because it moves prices and quantities in the same direction.
- M_{yA} and $M_{y\theta}$ do not depend on φ : money is neutral, and therefore the monetary policy rule has no effect on real quantities.
- $|M_{pA}|$ and $|M_{p\theta}|$ are decreasing functions of φ : when monetary policy is more aggressive, prices are stabilized (they respond less to shocks).

Additional answer-space for Question B4.

–Solution– See previous page.

PART C: Discussion (30 points)

Answer ALL questions of the following PART C. Each question is worth 10 points. Write your answers in words and include equations if needed or specified by the question. Record your answer to each question in the space below the question. If you need more space, use the ADDITIONAL SPACE at the end of the booklet, indicating clearly to which question(s) your answer pertains.

Below is an article published in The Guardian in July 2024 and entitled “Of the UK economy’s two possible endings, the happy one is more likely” and signed by Larry Elliott. Read carefully this article before answering the questions below. For each question, you should propose a structured answer and use the material we have seen in class.

Of the UK economy’s two possible endings, the happy one is more likely

Larry Elliott

The Guardian, 7th July 2024

Will the credits roll as Starmer wins big in 2029 or will Reeves be boxed in by her own fiscal rules and leave the economy in a worse state?



📷 Alternating storylines: Gwyneth Paltrow as Helen Quilley in the 1998 romantic comedy Sliding Doors. Photograph: Miramax/Paramount

Just as in the movie Sliding Doors, there are two endings to the story that began on Thursday with Labour’s victory in the 2024 election. One is happy, one is sad, and either is possible. What happens to the economy over the next five years will determine which one materialises.

Let’s start with the happy ending: the one where the credits roll as Keir Starmer wins big for a second time in 2029. The script for that one starts slowly. The scale of Labour’s win means it doesn’t have to rush things but has time to put in place the supply-side reforms, such as changes to planning rules, that will boost growth. Structural changes don’t bear fruit overnight but by the early 2030s it should be possible to notice the difference.

In the meantime the economy will benefit from a much-needed period of stability. Business investment was depressed during the period when there was uncertainty over whether the UK would leave the EU or not. It suffered another hit during the pandemic, and a third after Russia’s invasion of Ukraine. A chaotic period in which the UK had three prime ministers and five chancellors in a single parliament is now over.

PART C cont'd: Answer ALL questions from PART C

The poor showing of the SNP in the general election means the prospects of a second referendum on Scottish independence in the next five years are much diminished. Likewise, Starmer is unlikely to reconsider his decision to rule out rejoining the EU (or even the single market) with Reform UK breathing down Labour's neck in so many seats. As things stand, Britain looks a safer place for investment than France or Germany.

Much has been said about Labour's baleful economic inheritance, but growth is set to surprise on the upside in the coming months. Consumer spending power has been boosted in recent months because wages have been rising more rapidly than prices, but there tends to be a lag before households realise they are better off.

The election result will be the catalyst for higher consumer spending, and it will also prompt businesses to give the go-ahead for investment decisions put on hold in recent months.

The boost to growth will provide Rachel Reeves with a bit more scope in her autumn budget. She may also be tempted to make changes to the government's debt rules, which would provide her with some additional wriggle room.

It won't be a massive windfall, and the new chancellor says money will be tight. But she might have an extra £20bn to spend on public services.

Starmer and Reeves won't need to rush because the Tories will spend the post-election period at war with each other over the future direction of the party rather than in opposing what Labour is doing.

One big problem for Rishi Sunak in his period as prime minister was that the economic and political cycles became misaligned, with tough decisions, such as the freeze on tax allowances and thresholds, biting in the last years of the parliament. The prospects of a second Labour victory will be much enhanced if Reeves gets the unpopular stuff out of the way early and then lets up as the next election approaches.

The alternative ending is not nearly so cheerful. The gentle recovery in the economy runs out of steam as stubbornly high inflation in the service sector deters the Bank of England from cutting interest rates. Business investment continues to disappoint, while dearer fixed-rate mortgages and stagnant house prices make consumers cautious about spending money.

The situation cries out for the new government to boost public spending or cut taxes, but Reeves is boxed in by her own self-imposed fiscal rules. Public demands for better public services go unanswered, and disillusionment quickly sets in. Starmer and Reeves insist that their structural reforms will eventually pay dividends but, as was the case when Sunak tried the same line of argument, nobody is listening.

Labour's landslide victory was the result of a collapse in support for the Conservatives rather than a ringing endorsement of Starmer's offering. Never before has a party won so big a majority with such a modest share of the vote: 34%. Jeremy Corbyn won a bigger percentage of the vote when Labour was defeated in 2017.

Starmer and Reeves are on probation as far as Britain's sceptical and impatient voters are concerned, and they are deluding themselves if they think this is 1997 redux. The economy is weaker and public services are in a worse state. Sitting tight for two years is not an option.

It may take time for the Conservatives to recover from the pasting they have just taken, but there are alternative sources of opposition. The Green party threatens to pick up unhappy Labour voters from the left, while Reform UK could do the same from the right.

The Liberal Democrats will be attractive for those who want a closer relationship with the EU. The Tories won big in 2019 but last week lost more than 250 seats. Something similar could easily happen to Labour in 2029.

PART C cont'd: Answer ALL questions from PART C

So which ending is it going to be? On balance, the happy ending looks slightly more likely. That's not because the country is set for an economic boom – the UK has deep-seated problems that are not going to be solved overnight. But such was the desire of voters to be shot of the Conservatives that it will take time before they are trusted again.

The Tories need Labour to screw up the economy to have a chance of a quick comeback, but it is reasonable to assume that the next five years will be better than the previous five. That might not be saying much, but it should be enough.

End of the article

C1. Write a 10-bullet point summary of this article.

Record your answer to Question C1 in the space below. Do NOT use this space to record answers to any questions other than C1.

~~–Solution–~~ See next page.

Additional answer-space for Question C1.

–Solution–

1. Two Potential Outcomes: The article outlines two possible economic scenarios for the UK following Labour's victory in the 2024 election – one optimistic and one pessimistic, much like the movie Sliding Doors.
2. Optimistic Scenario: In the best case, Keir Starmer's government successfully implements supply-side reforms, including changes to planning rules that boost long-term growth. Economic stability and business investment are expected to improve as a result.
3. Economic Stability: Following years of uncertainty (Brexit, pandemic, Ukraine war), the UK enters a more stable period that supports consumer and business confidence, leading to higher economic growth.
4. Boost to Consumer Spending: Rising wages outpace inflation, giving consumers more disposable income and stimulating demand. This, in turn, supports business investment decisions that had been delayed in previous years.
5. Rachel Reeves' Budget: Labour's new Chancellor, Rachel Reeves, is expected to have some fiscal flexibility, possibly allowing an additional £20bn to invest in public services. However, there will be pressure to manage the budget carefully due to limited resources.
6. Political Advantage: The Conservative Party is likely to be preoccupied with internal divisions, making it difficult for them to mount a strong opposition to Labour's policies in the short term.
7. Pessimistic Scenario: In the less favorable outcome, inflation remains high in the service sector, preventing the Bank of England from lowering interest rates. High mortgage costs and stagnant house prices lead to lower consumer spending.
8. Fiscal Constraints: Labour's strict fiscal rules, which limit public spending, may prevent any significant fiscal stimulus. As a result, public services could continue to suffer from underinvestment, leading to disillusionment.
9. Voter Skepticism: Despite Labour's large victory in 2024, the public remains skeptical about their ability to deliver real change. Labour's mandate is seen as a rejection of the Conservatives rather than an enthusiastic endorsement of Starmer's agenda.
10. Future Risks: Labour's political dominance is not guaranteed, as opposition parties (Greens, Reform UK, Liberal Democrats) could siphon off voters if economic conditions worsen. Labour's ability to manage the economy will be critical to its long-term political success.

PART C cont'd: Answer ALL questions from PART C

C2. It is written “*In the meantime the economy will benefit from a much-needed period of stability. Business investment was depressed during the period when there was uncertainty over whether the UK would leave the EU or not*” and “*As things stand, Britain looks a safer place for investment than France or Germany*”. Can you make sense of these sentences in a “UV” model of Lecture 2 on Business Cycles?

Record your answer to Question C2 in the space below. Do NOT use this space to record answers to any questions other than C2.

–Solution–

- These sentences refer to the changes of investors willingness to invest along the business cycle.
- In the “UV” model, this was modelled as a shift in the continuation value of investment. In that model, preferences are

$$U(c, \ell) + \theta V(x)$$

where c is consumption, ℓ labour and x investment. V is the continuation value of investment, and “*As things stand, Britain looks a safer place for investment than France or Germany*” can be translated as an increase in θ , which may lead to an increase in economic activity.

- Whether or not a demand shock like θ is expansionary depend on the model of the economy. In the RBC model, it will increase x but decrease c . In a model with sticky prices (very much like an IS-LM model) or in a “Gains From Trade” model, it can raise at the same time c and x (and ℓ).

PART C cont'd: Answer ALL questions from PART C

C3. It is written “*The alternative ending is not nearly so cheerful. The gentle recovery in the economy runs out of steam as stubbornly high inflation in the service sector deters the Bank of England from cutting interest rates*”. Can you make sense of these sentences?

Record your answer to Question C3 in the space below. Do NOT use this space to record answers to any questions other than C3.

–Solution–

- Monetary policy is facing a trade off between short run growth and inflation.
- By setting the nominal interest rate, the Central Bank has an impact on the economy.
- Lax monetary policy, by cutting the interest rate, is expected to foster short run growth, at the cost of higher inflation.
- In the pessimistic scenario of the article, because of inflationary pressures in the service sector in the UK, the BoE cannot follow a lax monetary policy, as this will exacerbate inflationary pressures.

If you need extra space for a question, you may use the Additional Space on this page should you require it, indicating clearly to which question(s) your answers pertain.

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End of Exam Checklist:

Please ensure that:

- You have correctly written your student number and name on the cover sheet (page 1).
- Your answer-booklet is complete and not damaged.

The remainder of this page has been left blank intentionally. Do not use the space on this page to record your answers to questions in this exam.